

LF11c:03:Techniken der Datenaufbereitung für die Ausbildung zur Fachinformatikerin

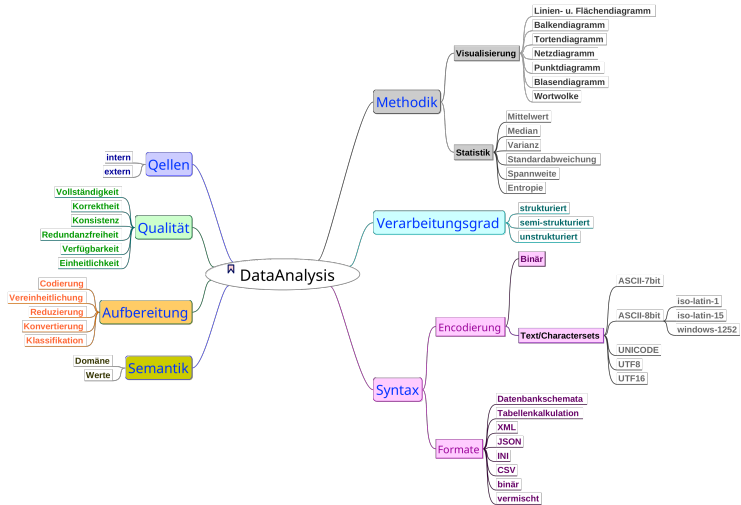
Karsten Reincke

Gewerbliche Schulen Dillenburg

7. September 2026

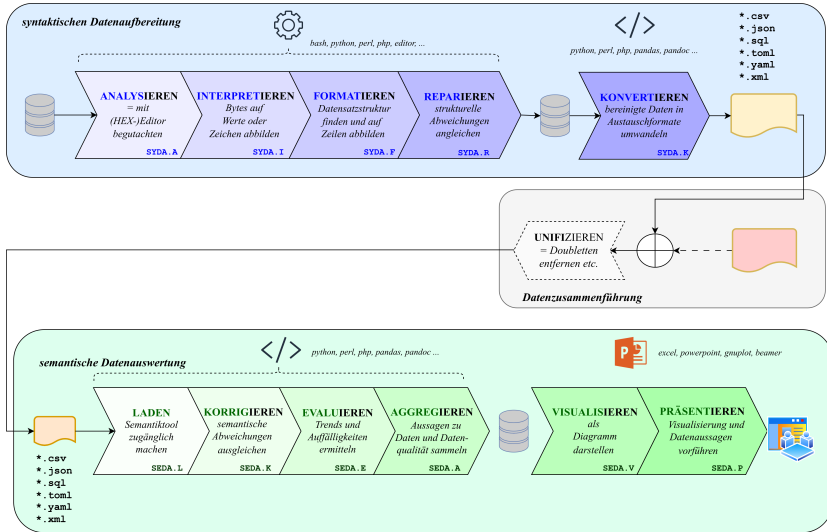
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LF11c:03:Datenanalyse:Dimensionen

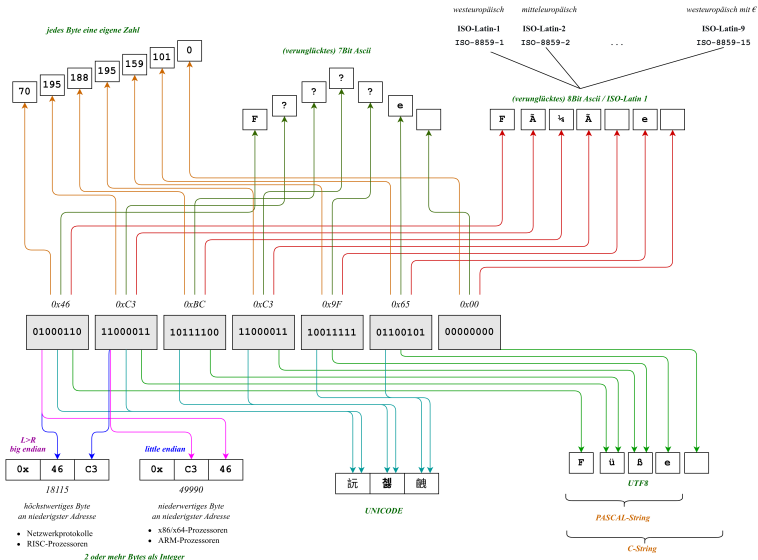


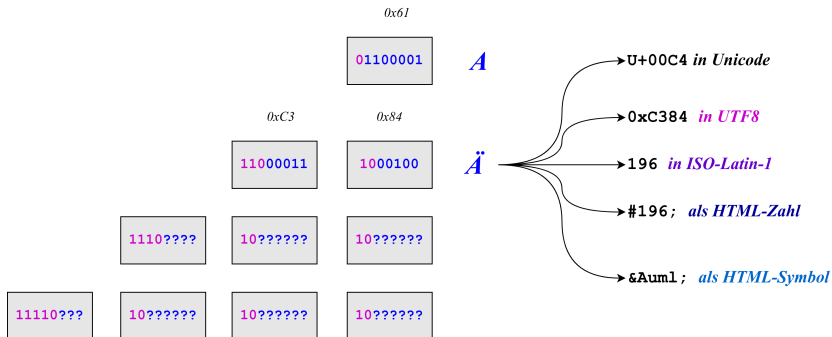


LF11c:03:Datenanalyse:Phasen

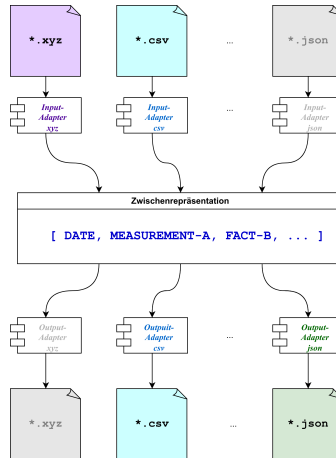


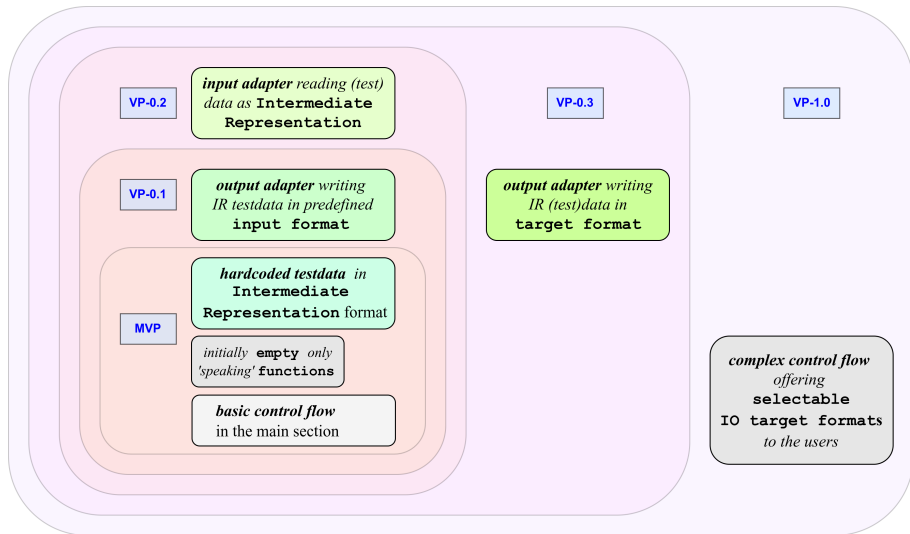
LF11c:03:Datenaufbereitung:Byte-Interpretation





LF11c:03:Datenaufbereitung:Konverter:Architektur







Initial erhalten:

```
1 2026-01-01,08:29,16:31,08:02,2026-01-02,08:29,16:32,08:03,2026-01-03,08
2 :29,16:33,08:04,2026-01-04,08:28,16:34,08:05,20260103,08:28,16:34,08:10
```

SYDA.A - SYDA.F: Reformatiert und Komma nach Datensatz entfernt:

```
1 2026-01-01,08:29,16:31,08:02
2 2026-01-02,08:29,16:32,08:03
3 2026-01-03,08:29,16:33,08:04
4 2026-01-04,08:28,16:34,08:05
5 20260103,08:28,16:34,08:10
```

SYDA.K: 5. Datensatz entfernt, weil falsches Datumsformat + Dublette:

```
1 2026-01-01,08:29,16:31,08:02
2 2026-01-02,08:29,16:32,08:03
3 2026-01-03,08:29,16:33,08:04
4 2026-01-04,08:28,16:34,08:05
```



LF11c:03:Datenaufbereitung::dsp-beta:MVP

```

1  # (C) 2026 K.Reincke: proTirone snippet [CC-BY-4.0]
2
3  # converts a csv file of day data into the respective ini file: MVP
4
5  # intermediate representation and hard-coded testdata
6  dd_hard_coded=[
7  { "day":"2026-01-01", "sunrise": "08:29","sunset": "16:31","durance": "08:02"},
8  { "day":"2026-01-02", "sunrise": "08:29","sunset": "16:32","durance": "08:03"}]
9
10 # adapters to read daydata from a file:
11 def fill_from_csv(filename) :
12     print(f"reading from csv file <{filename}>")
13     daydata=[]
14     return daydata
15
16 # adapters to write daydata into a file
17 def write_as_csv(filename,daydata):
18     print(f"writing into csv file <{filename}>")
19
20 # main
21 dd_csv_file="res.lf/daydata.csv"
22
23 daydata=dd_hard_coded
24 if (daydata):
25     write_as_csv(dd_csv_file,daydata)
26

```



LF11c:03:Datenaufbereitung::dsp-beta:VP-0.1

```
1  # (C) 2026 K.Reincke: proTirone snippet [CC-BY-4.0]
2
3  # converts a csv file of day data into the respective ini file: VP-0.1
4
5  # intermediate representation and hard-coded testdata
6  dd_hard_coded=[
7  { "day":"2026-01-01", "sunrise": "08:29","sunset": "16:31","durance": "08:02"},
8  { "day":"2026-01-02", "sunrise": "08:29","sunset": "16:32","durance": "08:03"}]
9
10 # adapters to read daydata from a file
11 def fill_from_csv(filename) :
12     print(f"reading from csv file <{filename}>")
13     daydata=[]
14     return daydata
15
16 # adapters to write daydata into a file
17 def write_as_csv(filename,daydata):
18     print(f"writing into csv file <{filename}>")
19     ouf=open(filename,"w")
20     # keep it simple: let the program crash if filename does not exist
21     for ds in daydata:
22         ouf.write(f"{ds["day"]},{ds["sunrise"]},{ds["sunset"]},{ds["durance"]}\n")
23     return ouf.close()
24
25 # main
26 dd_csv_file="res.lf/daydata.csv"
27
28 daydata=dd_hard_coded
29 if (daydata):
30     write_as_csv(dd_csv_file,daydata)
```

LF11c:03:Datenaufbereitung::dsp-beta:Adapter

```

1  # (C) 2026 K.Reincke: proTirone snippet [CC-BY-4.0]
2  # functions to write and to convert daydata (dsp-beta) csv files
3
4  # (1) adapters to read daydata from a file
5  def fill_from_csv(filename) :
6      print(f"reading from csv file <{filename}>")
7      daydata=[]
8      inf=open(filename,"r")
9      # keep it simple: let the program crash if filename does not exist
10     for line in inf:
11         csv_str=line.strip()
12         (dy,se,st,de)=csv_str.split(',')
13         daydata.append({"day":str(dy).strip(),"sunrise":str(se).strip(),
14                         "sunset":str(st).strip(),"durance":str(de).strip()})
15     return daydata
16
17 # (1) adapters to write daydata into a file
18 def write_as_csv(filename,daydata):
19     print(f"converting as csv file <{filename}>")
20     ouf=open(filename,"w")
21     # keep it simple: let the program crash if filename does not exist
22     for ds in daydata:
23         ouf.write(f"{ds["day"]},{ds["sunrise"]},{ds["sunset"]},{ds["durance"]}\n")
24     return ouf.close()
25
26 def write_as_ini(filename,daydata):
27     print(f"converting as ini file <{filename}>")
28     ouf=open(filename,"w")
29     # keep it simple: let the program crash if filename does not exist
30     for ds in daydata:
31         ouf.write(f"[ {ds["day"]} ]\nSunrise={ds["sunrise"]}\nSunset={ds["sunset"]}\nDurance={ds["durance"]}\n")
32     return ouf.close()

```



LF11c:03:Datenaufbereitung::dsp-beta:VP-1.0

```
1  # (C) 2026 K.Reincke: proTirone snippet [CC-BY-4.0]
2  # converts a csv file of day data into the respective ini file: VP-1.0
3
4  import sys
5  import mod_dsp_beta_conv_adapters as adapters
6
7  # intermediate representation and hard-coded testdata
8  dd_hard_coded=(
9  { "day":"2026-01-01", "sunrise": "08:29","sunset": "16:31","durance": "08:02"},
10 { "day":"2026-01-02", "sunrise": "08:29","sunset": "16:32","durance": "08:03"})
11
12 dd_in_file="dsp.lf/dsp.beta.csv"
13 dd_out_format="csv"
14
15 if ((len(sys.argv)>1)) :
16     match sys.argv[1]:
17         case "INI" | "ini": dd_out_format="ini"
18         case "test": dd_in_file="dsp.lf/dsp.beta-t.csv"
19 if ((len(sys.argv)>2)) :
20     match sys.argv[2]:
21         case "INI" | "ini": dd_out_format="ini"
22
23 # main
24
25 daydata=adapters.fill_from_csv(dd_in_file)
26 #daydata=dd_hard_coded
27
28 if (daydata):
29     if(dd_out_format=="csv"):
30         adapters.write_as_csv("res.lf/daydata.csv",daydata)
31     else:
32         adapters.write_as_ini("res.lf/daydata.ini",daydata)
```



Herzlichen Dank für Ihre Aufmerksamkeit!



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